Quality Report

PIX4Dmatic v2.5.1

Camera	DJI_FC300S_3.6_4000x3000
Area covered	1.637 ha
Average GSD	2.0 cm
Project CRS	WGS 84 / UTM zone 17N + EGM96 height - EPSG:32617+5773 [EGM96]
Dense point count	3,472,521

Quality check

Matches	Median of 43 matches per calibrated image Very low number of matches (median < 100).	!
Dataset	98.61% calibrated (71/72 images), 3 calibration blocks	✓
Camera optimization	1.44% relative difference between initial and optimized internal camera parameters	✓
MTPs	4 MTPs, RMS reprojection error 0.2 px / Sigma 0.2	✓
ATPs	1204 ATPs	

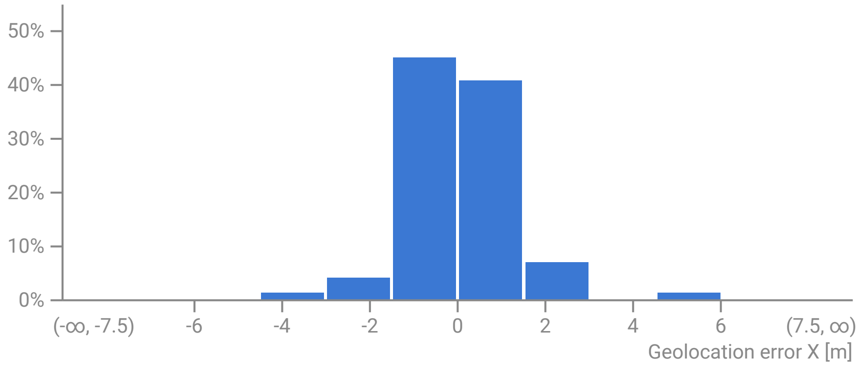
Cameras

Internal camera parameters

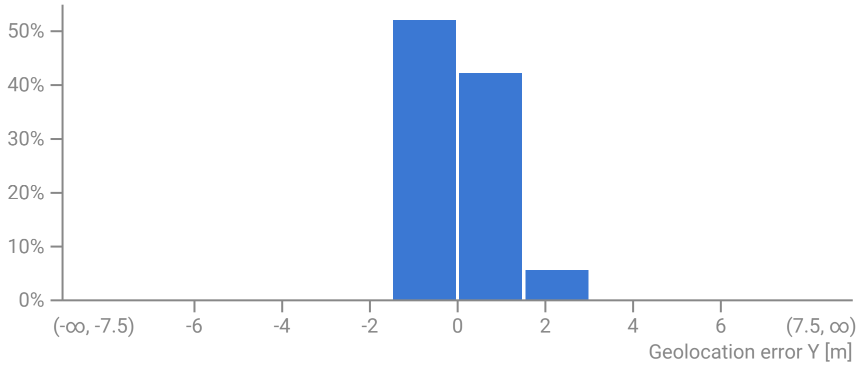
DJI_FC300S_3.6_4000x3000. Sensor dimensions: 6.317 mm x 4.738 mm

	Focal length	Principal point x	Principal point y	R1	R2	R3	T1	T2
Initial	2285.7 px 3.61 mm	2000.0 px 3.159 mm	1500.0 px 2.369 mm	-0.1324380	0.1110560	-0.0158256	0.0001102	0.0001142
Optimized	2318.6 px 3.662 mm	2024.0 px 3.197 mm	1500.8 px 2.37 mm	-0.0200061	-0.0141641	0.0346922	0.0003631	0.0004905
Uncertainties (Sigma)	80.8 px 0.128 mm	39.8 px 0.063 mm	51.4 px 0.081 mm	0.0000684	0.0002880	0.0001340	0.0000004	0.0000007

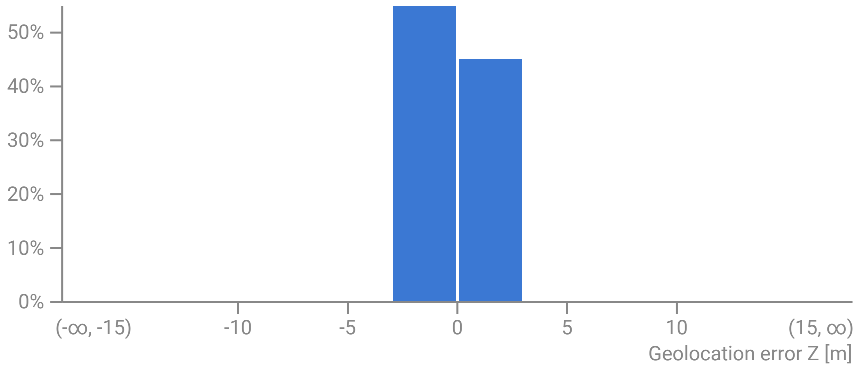
Absolute geolocation variance



	Geolocation error X [m]
Mean	0.009
Median	-0.002
Sigma	1.227
RMS	1.227



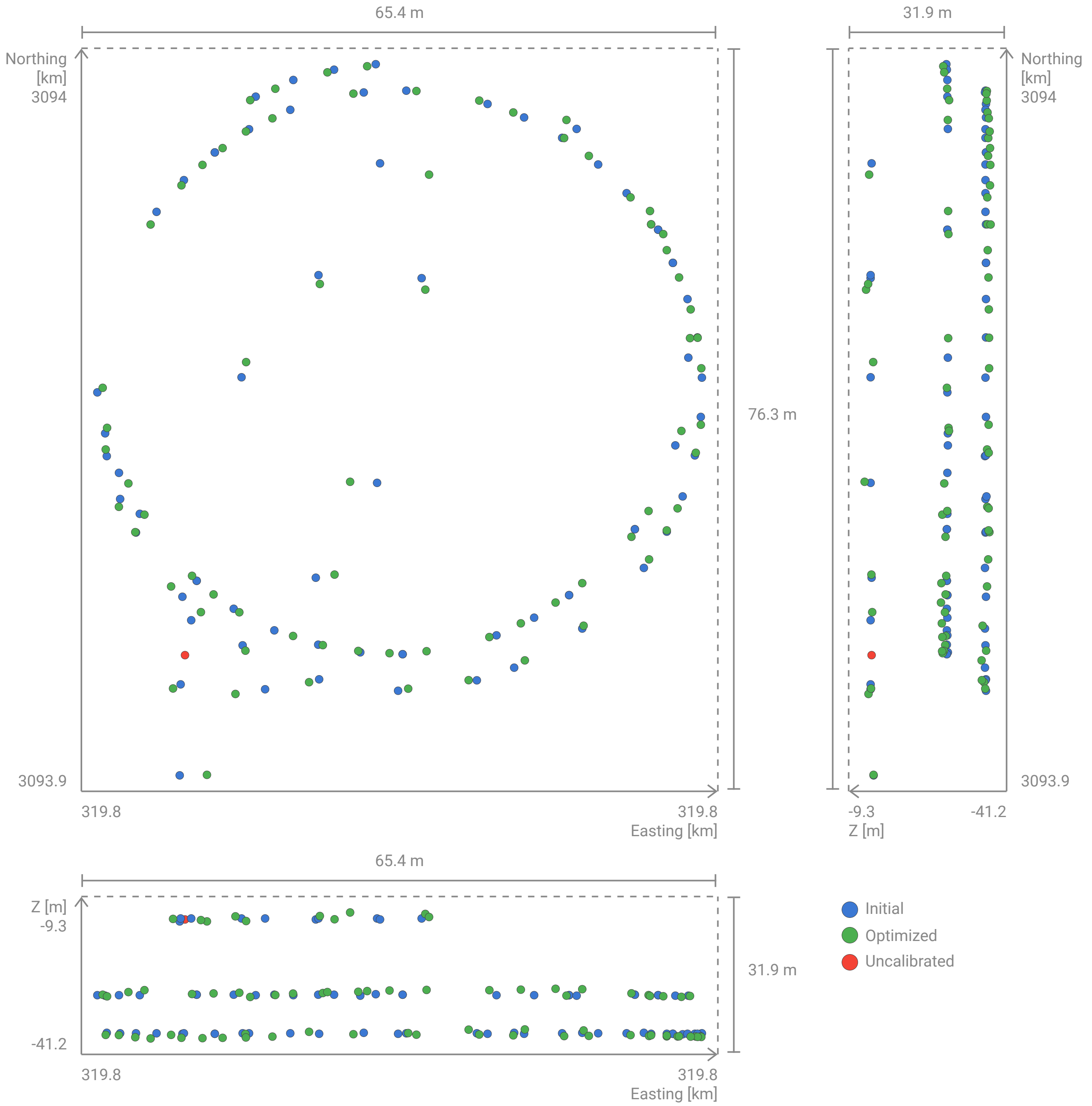
	Geolocation error Y [m]
Mean	-0.002
Median	-0.033
Sigma	0.828
RMS	0.828



	Geolocation error Z [m]
Mean	0.004
Median	-0.135
Sigma	0.595
RMS	0.595

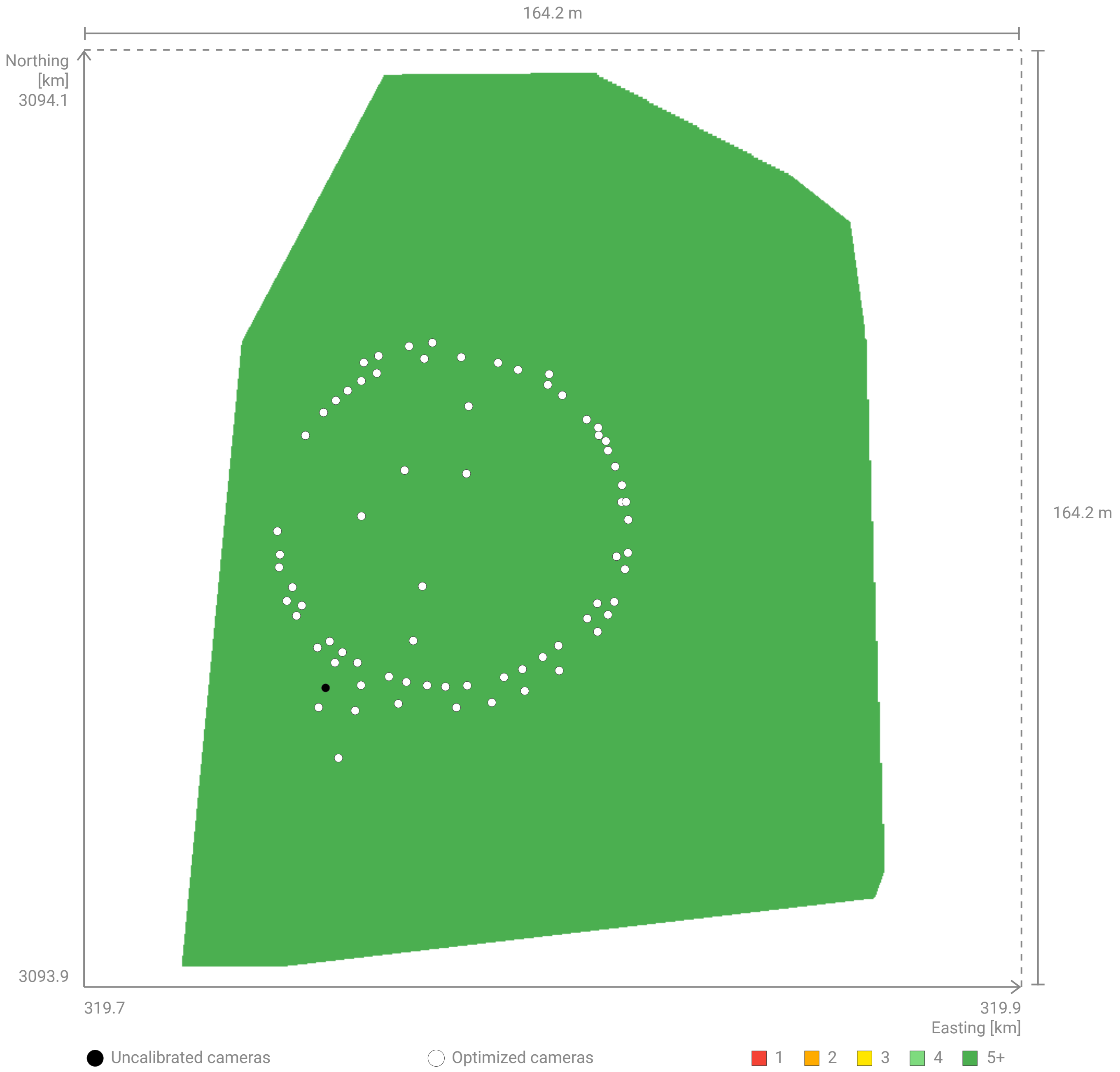
The geolocation error is the difference between the initial and computed camera positions. Plots show the per-axis distributions of geolocation errors across the cameras. Large positive and negative errors are denoted with the orange bins. Note that the image geolocation errors do not correspond to the accuracy of the observed 3D points.

Camera positions



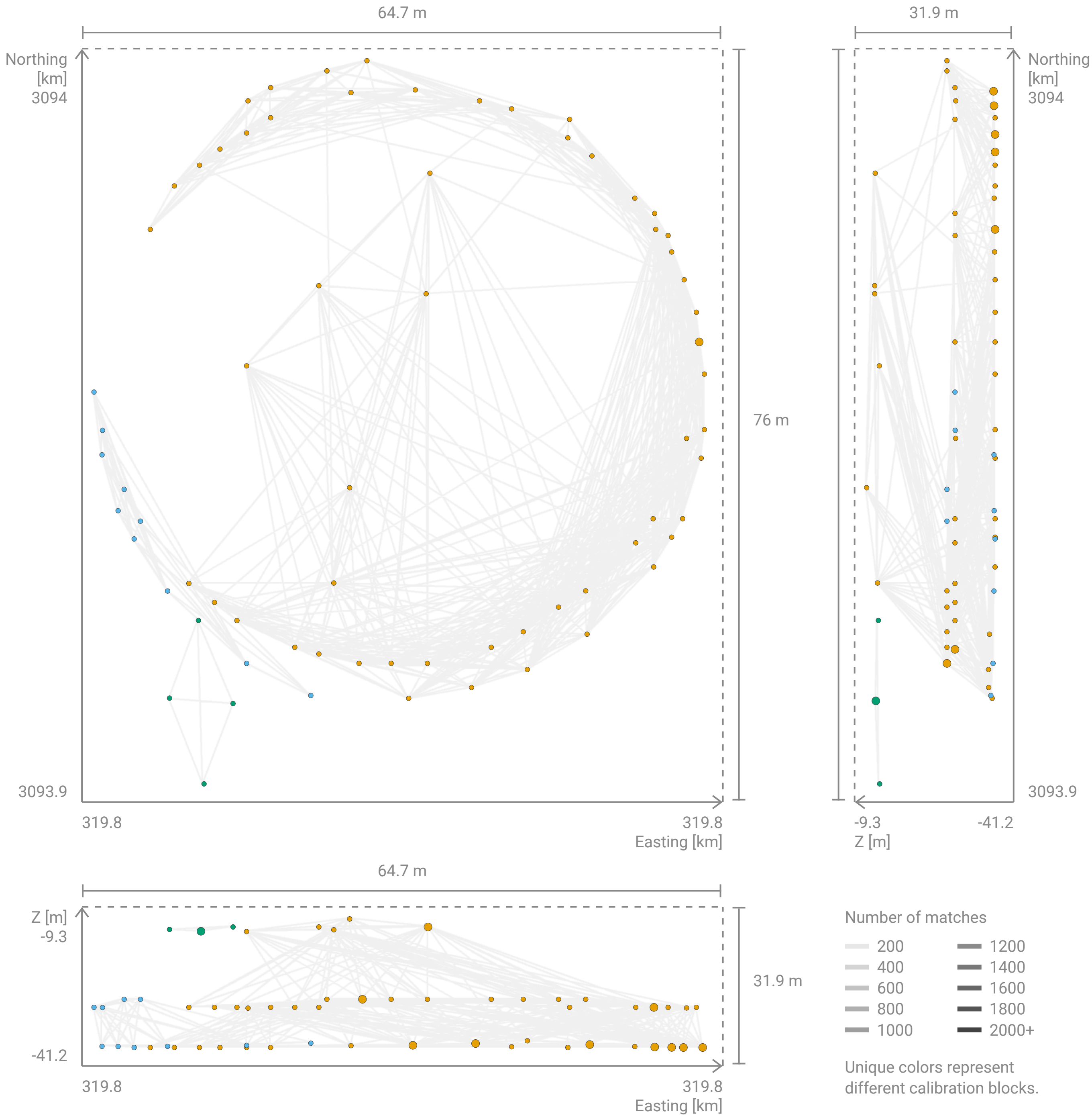
Initial and computed camera positions.

Overlap



This graph shows the number of overlapping images for each pixel of the DSM preview. For precise 3D modeling and mapping applications, the overlap should be in green, i.e. each pixel should be visible in more than 5 images.

2D Keypoint matches



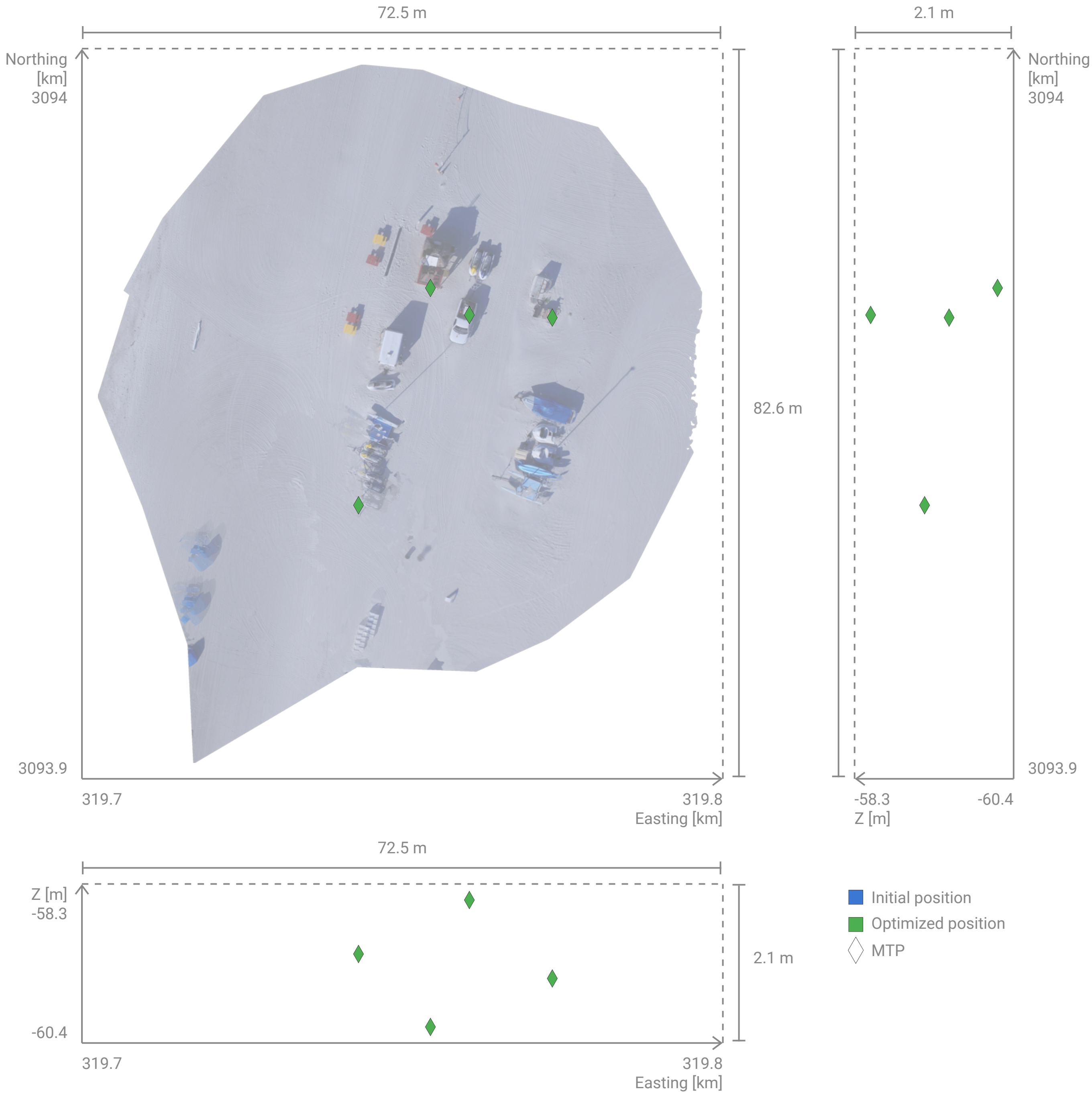
Computed camera positions with links between matched cameras. The opacity of the links indicates the number of matched 2D keypoints between the cameras. Near-transparent links indicate weak links and require manual tie points or more cameras. To improve visibility, camera positions may be slightly shifted and multiple cameras may be grouped into a single point on the plot. Group of multiple cameras is indicated by a larger point on the plot.

Tie points

Manual tie points (MTPs)

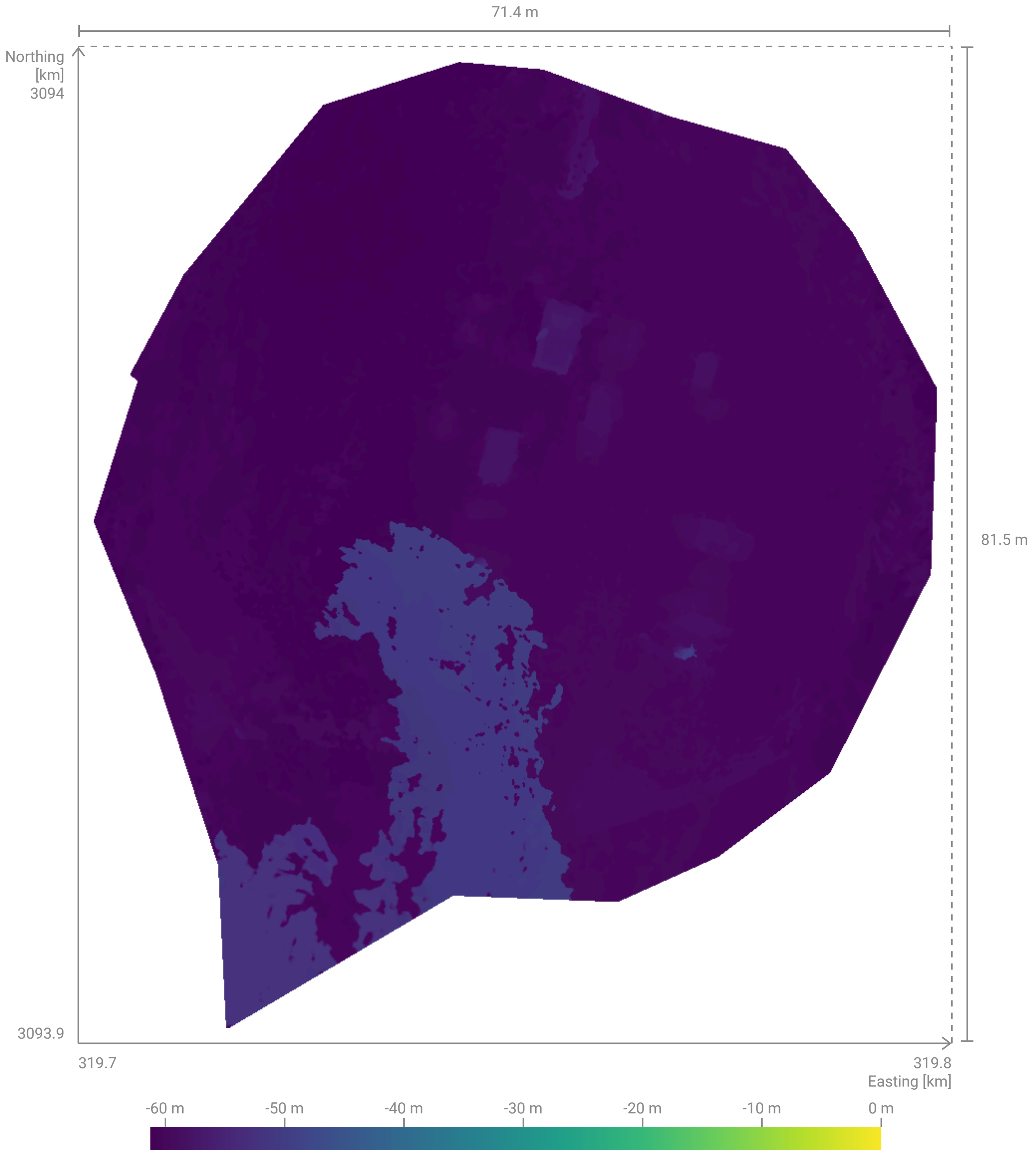
Label	Reprojection error [px]	Verified/Marked
MTP7	0.2	9/9
MTP8	0.1	10/10
MTP10	0.4	10/10
MTP15	0.0	8/8
Min	0.0	
Max	0.4	
Mean	0.2	
Median	0.1	
Sigma	0.2	
RMS	0.2	

Tie point positions



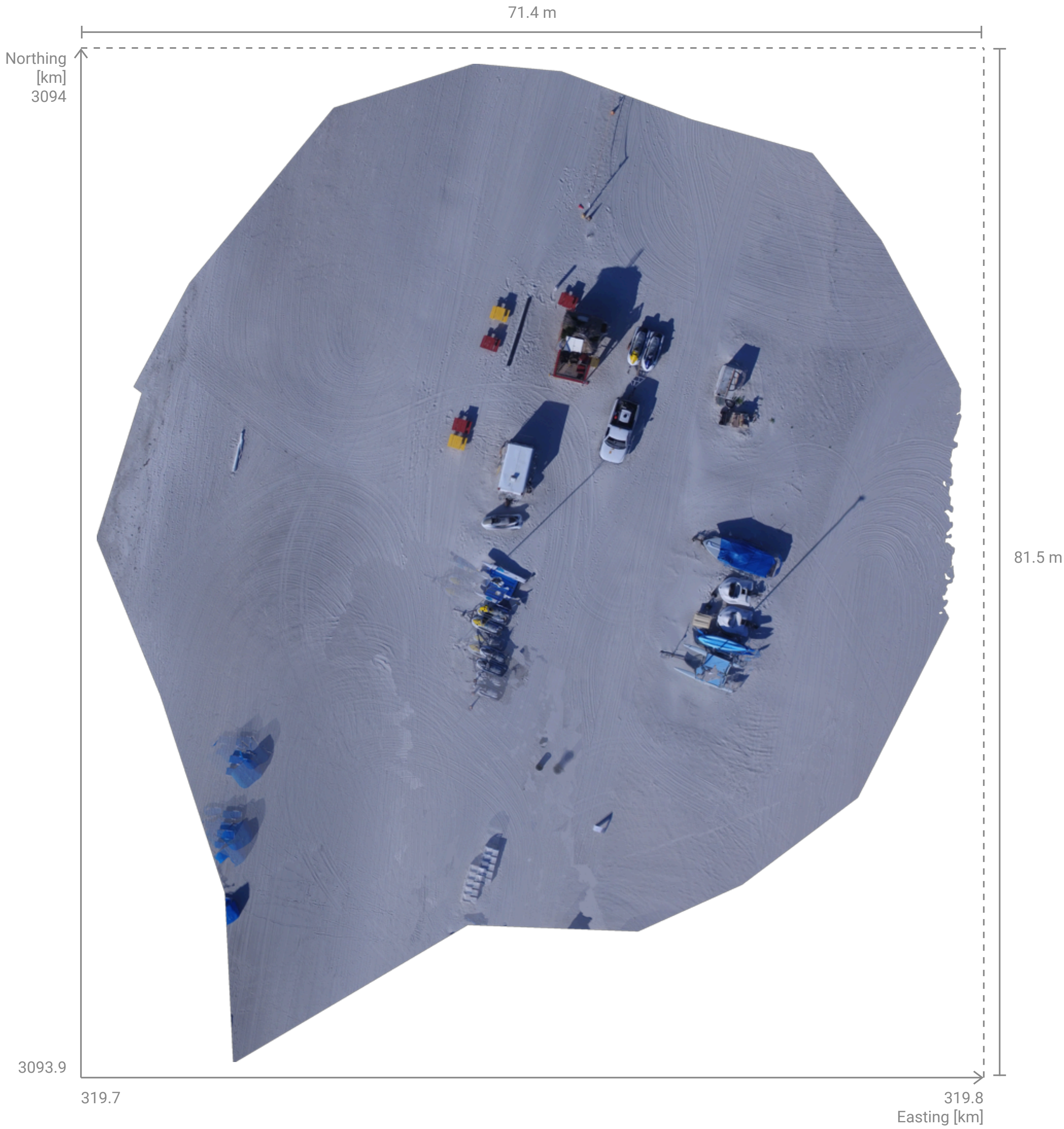
Initial or optimized position of tie points.

DSM



Resolution: 2.0 cm/px
Image resolution: 846x970 px

Orthomosaic



Resolution: 2.0 cm/px
Image resolution: 846x970 px

Hardware & Settings

System information

CPU	13th Gen Intel(R) Core(TM) i9-13900H, cpus=1, threads=20
RAM	15.63 GB
GPU	Intel Intel(R) Iris(R) Xe Graphics (Driver: 4.1.0 - Build 32.0.101.5768)
Operating system	Windows 11

Coordinate reference systems

Image coordinate reference system	WGS 84 + EGM96 height - EPSG:4326+5773 [EGM96]
Project coordinate reference system	WGS 84 / UTM zone 17N + EGM96 height - EPSG:32617+5773 [EGM96]

Processing settings

Calibration — Completed	Calibration: Reoptimization — Completed	Dense point cloud — Completed
Template: Large scale and corridor **Pipeline:** Scalable standard **Image scale:** 1/8 **Internals confidence:** Low **Externals confidence:** Standard **Simultaneous camera internals and MTP/GCP optimization:** Enabled **Max. extracted keypoints:** 10000 **Use automatic ITPs:** Disabled	**Rematch:** Disabled **Rematch template:** Large scale and corridor **Camera internals:** Optimized internals **Internals confidence:** Low **Externals confidence:** Standard **Simultaneous camera internals and MTP/GCP optimization:** Enabled	**Algorithm:** Hardware accelerated **Image scale:** 1/2 **Density:** Optimal **Min. number of matches:** 3 **Multiscale:** Enabled **Noise filter:** Disabled **Sky filter:** Disabled **Mask-aware:** Disabled
16s	1s	3m 6s

Mesh — Completed	DSM — Completed	Orthomosaic — Completed
Input point cloud: Dense point cloud **Quality:** Detailed **Template:** Nadir **Model simplification:** Strong **Deghosting:** Weak **Polygon-aware:** Disabled **Sky mask:** Disabled **Mask-aware:** Disabled **Interior improvement:** Disabled	**Input point cloud:** Dense point cloud **Interpolation:** Interpolate holes **Resolution:** 2.0 cm/px **Surface smoothing:** 8 px **Polygon-aware:** Disabled **Mask-aware:** Disabled	**Resolution:** 2.0 cm/px **Algorithm:** Hardware accelerated **Oblique:** Disabled **Blending algorithm:** Full **Mask-aware:** Disabled
4m 37s	10s	15s